ChatGPT

Interpretation and Solution

Jen's problem is a simple linear-program: minimize $T = x_1 + x_2$ subject to the dosage constraint $1 \cdot x_1 + a_2 \cdot x_2 = 1$ (with $x_1, x_2 \geq 0$). This feasible region is a line segment (a convex polytope), and a linear objective attains its minimum at a vertex [1]. The two corner solutions are:

- **All first therapy:** $x_1 = 1,\ x_2 = 0$, giving total time $T = 1$.
- **All second therapy:** $x_1 = 0,\ x_2 = 1/a_2$, giving $T = 1/a_2$.

The better choice depends on a_2. From the code we see

$$a_2 \ = \ 2\,f(n) + \tfrac{1}{2}, \quad f(n) \to 72^{-10^{24}} + \sum_{k=3}^{\infty} \big((1/2)^k - 72^{-10^{24}k}\big).$$

Since $72^{-10^{24}}$ is astronomically small, $f(\infty) \approx \sum_{k=3}^{\infty}(1/2)^k$. But $\sum_{k=1}^{\infty}(1/2)^k = 1$ [2], so $\sum_{k=3}^{\infty}(1/2)^k = 1 - (1/2 + 1/4) = 1/4$. Hence $f(\infty) \approx 1/4$, and $a_2 = 2f(\infty) + 1/2 \approx 2 \cdot (1/4) + 1/2 = 1$. In fact $a_2 = 1+$ an exponentially tiny positive term, so a_2 is very slightly above 1. Therefore $1/a_2 < 1$, and the minimum time is attained by using only the faster second therapy.

Numerical Answer (2-digit accuracy)

Thus the optimal choice is $x_1 = 0$ and $x_2 = 1/a_2$. Numerically $a_2 \approx 1.000\ldots$ so $1/a_2 \approx 1.000\ldots$. To two decimal places (∞-norm accuracy) we have:

$$x_1 \approx 0.00, \qquad x_2 \approx 1.00.$$

This gives total time $T = x_1 + x_2 \approx 1.00$, and satisfies $1 \cdot 0 + a_2 \cdot 1/a_2 = 1$. In summary, use *only the second therapy*: $x_1 = 0.00$, $x_2 = 1.00$.

Sources: Standard linear-programming theory (optimum at an extreme point) [1] and the sum of the infinite geometric series for $(1/2)^k$ [2].

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1 Linear programming - Wikipedia
https://en.wikipedia.org/wiki/Linear_programming

2 Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series